

requests such as video playback. Media Foundation is meant to replace DirectShow of old and handles all multimedia requests from applications such as audio and video editors, music and movie players, and so on. It essentially provides a protected path to media formats, so much so that a DRM-flagged content piece can potentially refuse to be scanned for a system threat (like a virus scan). Vista offers media to employ such protected paths which would only be accessible to plugins that have the proper authority to access the program.

Thus, a video player could potentially refuse to not only play a DRM protected file, but might also refuse to run a plugin, if it is not signed according to specification and requirement. This has been done to avoid hacks to HD content which might be either internal or via an outside attack through a third-party plugin. Apart from media protection, Media Foundation also introduces support for hardware acceleration of popular codecs. Windows XP provides such acceleration through the DirectX Video Acceleration for MPEG-2 content (generally used in DVDs). Media Foundation extends this codec support to the H.264 and VC-1 codecs. Videos encoded using these would thus play smoother, given the appropriate hardware support.

Windows Vista also refreshes the audio architecture as the Universal Audio Architecture. This new structure provides hi-fidelity support for professional creation of audio, apart from providing separate mixers to disparate audio streams from multiple software sources. Hi-fidelity sound also requires an architecture which offers the support needed in terms of greater bandwidth, such that latencies while working on the hi-fidelity sound streams are reduced. UAA does just that, especially at higher bitrates. It is also meant to allow simultaneously working with multiple sources of audio, be it playback or recording. Under UAA, all audio streams should now be high-definition playing at a frequency of at least 96 kHz at 24-bit; audio support should also extend to 5.1 channels wherever possible.